Nuclear and Risk Engineering

	Normative 4-year Degree
Major	Nuclear and Risk Engineering 核子及風險工程
Degree / Award Title	Bachelor of Engineering in Nuclear and Risk Engineering 工學士(核子及風險工程)
Offering Academic Unit	This Major is jointly offered by: Department of Mechanical Engineering Department of Physics Department of Chemistry Department of Architecture and Civil Engineering Department of Electrical Engineering Department of Systems Engineering
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business	12

	Organisations Area 3: Science and Technology Visit the GE website for more course information.	
College-specified Courses	MA1200 Calculus and Basic Linear Algebra I *Or* MA1300 Enhanced Calculus and Linear Algebra I	3
	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	MNE2066 Engineers in Society	3

College / School Requirements (6 credit units) — Catalogue Term : Semester A 2025/26

Course Code	Course Title	Credit Units
CS1302	Introduction to Computer Programming	3
PHY1101	Introductory Classical Mechanics	3

Major Requirements (81 credit units) — Catalogue Term : Semester A 2026/27

1. Core Courses (72 credit units)

Course Code	Course Title	Credit Units	Remarks
MA2001	Multi-variable Calculus and Linear Algebra	3	
MA3001	Differential Equations	3	
MNE2020	Engineering Workshop Practice	0	
MNE2029	Electrical and Electronic Principles I	3	
MNE2036	Engineering Computing	3	
MNE2109	Engineering Mechanics	3	
MNE2110	Engineering Materials	3	
MNE2112	Thermodynamics	3	
MNE2116	Engineering Graphics	0	# Department-specified Course for top 40% majors allocation
MNE3049	Control Principles	3	
MNE3107	Principles of Nuclear Engineering	3	
MNE3118	Mechanics of Materials	3	
MNE3119	Manufacturing Technology	3	
MNE3121	Heat Transfer	3	
MNE3122	Fluid Mechanics	3	
MNE4105	Nuclear Reactor Safety	3	
MNE4109	Reliability Engineering and Risk Analysis	3	

MNE4112	Nuclear Materials	3	
MNE4118	Project (Individual)	6	A course to fulfil the internship/ consultancy project/research project requirement
MNE4132	Quantitative Risk Management	3	
MNE4231	Nuclear Reactor Physics	3	
PHY1202	General Physics II	3	# Department-specified Course for top 40% majors allocation
PHY3210	Modern Physics for Nuclear Technology	3	
PHY3230	Nuclear Radiation and Measurements	3	
PHY3275	Radiation Protection and Dosimetry	3	

2. Electives (9 credit units)

Course Code	Course Title	Credit Units	Remarks
CA4644	Wind and Earthquake Engineering	3	Pre-cursor waiver given by Department of Architecture and Civil Engineering.
CA4737	Fire Science and Modelling	3	Pre-cursor waiver given by Department of Architecture and Civil Engineering.
CHEM3038A	Environmental Sampling and Risk Assessment	3	Students are advised to take the course CHEM1200 Discovery in Biology before taking this course.
FS2001	Workshop-based Study in Science and Engineering	3	
MNE2113	Aircraft System Design	3	
MNE2114	Experimental Sustainable Engineering Techniques	3	
MNE3007	CAD/CAM	3	@
MNE3046	Automation Technology	3	#
MNE3109	Hazard Effect Management Process	3	
MNE3110	Safety Engineering Design	3	
MNE3123	Internship in Engineering	3	
MNE3125	Fluid Mechanics for Energy-Related Applications	3	
MNE3126	Methodologies for Solving Complex Engineering Problems	3	
MNE3127	Electron Microscopy	3	
MNE4005	Finite Element Analysis	3	&

MNE4010	Dynamics and Vibrations	3	
MNE4047	Directed Studies	3	
MNE4108	Nuclear Reactor Engineering	3	
MNE4120	Bioinspired Composites: Design, Mechanics and Manufacturing	3	
MNE4121	Machine Learning and Quantum Computation	3	
MNE4122	Introduction to Microfluidics	3	
MNE4123	Renewable and Sustainable Energy Systems	3	
MNE4125	Principles and Transport Process Fundamentals of Semiconductor Manufacturing	3	
MNE4130	Introduction to Multiphase Flow Science	3	
MNE4131	Nuclear Reactor Design	3	
MNE5112	Mechanical Design with Advanced Material & Additive Manufacturing	3	Only for final year undergraduate students (with quota restrictions)
MSE3169			
MSE3171	Materials Characterization	3	
PHY4230	Radiation Safety	3	
PHY4232	Radiotherapy Physics	3	
PHY4233	Imaging Physics	3	
PHY4274	Radiation Biophysics	3	
PHY4275	Radiological Physics and Dosimetry	3	
PHY4283	Physics in Medicine	3	
SYE3102	Quality Engineering	3	
SYE4064	Reliability Engineering	3	
SYE4103	Decision Analysis and Risk Management	3	

@ Course that would contribute towards the area of 'Design and Manufacturing' for Mechanical Engineering discipline of Scheme A training of HKIE.
# Course that would contribute towards the area of 'Automatic & Control Systems' for Mechanical Engineering discipline of Scheme A training of HKIE.
& Course that would contribute towards the area of 'Solid Mechanics' for Mechanical Engineering discipline of Scheme A training of HKIE.

3. Optional Courses

Course Code	Course Title	Credit Units	Remarks
FS2002	Industrial Attachment Scheme	3	Internship (minimum 6 weeks)
FS3002	Industrial Attachment Scheme	3	Internship (minimum 6 weeks)

FS4001	Co-operative Education Scheme (CES)	8	Internship (8 - 12 months)
FS4002	Industrial Attachment Scheme	3	Internship (minimum 6 weeks)
FS4005	Overseas Internship Scheme	3	Internship (9 - 13 weeks)